

BITS, PILANI – K. K. BIRLA GOA CAMPUS

Theory of Computation

(CS F51)

Lecture No. 13



Non Regular Languages

Observe that for a *regular language*,

1. Accepted by a *FSA*

The amount of memory(states) that is needed to determine whether or not a given string is in the language is *finite* and independent of the length of the string

2. Generated by an *regular expression*

If the language consists of an infinite number of strings, then this language should contain infinite subsets having a fairly *repetitive structure*.

Non Regular Languages

Intuitive candidates for *non regular languages*:

1. $L = \{w \in \{0, 1\}^* : w = 0^n 1^n, n \geq 0\}$

This language should be nonregular, because it seems unlikely that a FSA can remember how many 0s it has seen when it has reached the border between the 0's and the 1's.

2. $L = \{1^p : p \text{ is a prime}\}$

Because the prime numbers do not seem to have any repetitive structure that can be used by a regular expression

Non Regular Languages

We will establish a property that all *regular languages* must possess.

This property is called the *pumping lemma*.

If a language does not have this property, then it must be *nonregular*.

Pumping Lemma for *Regular* Languages

Let L be a *regular* language.

Then there exists an integer $n \geq 1$, called the *pumping length*, such that the following holds:

For *every string* $w \in L$, with $|w| \geq n$, *can be rewritten/decomposed* as $w = xyz$, such that

1. $y \neq \epsilon$

2. $|xy| \leq n$, and

3. $xy^iz \in L$ *for each* $i \geq 0$

By replacing the portion y in w by **zero or more copies** of it, the resulting string is still in the language L .

First non regular language

First non regular language

$L = \{0^n 1^n : n \geq 0\}$ is not a regular language

Suppose L was regular

Let $n \geq 1$ be the pumping length as in Pumping Lemma

Consider the string $w = 0^n 1^n$

Key is to choose the right string.

Then $w \in L$ and $|w| = 2n \geq n$

For example $w = 0^{n/2} 1^{n/2}$ will not help much

Hence, by the pumping lemma, w can be rewritten/decomposed as

$w = xyz$, where $y \neq \epsilon$, $|xy| \leq n$, and $xy^i z \in L$ for each $i \geq 0$

Consider any decomposition of w as $w = xyz$, where $y \neq \epsilon$, $|xy| \leq n$, and $xy^i z \in L$ for each $i \geq 0$

Since $|xy| \leq n$, the string y contains only 0's

Moreover, since $y \neq \epsilon$, y contains at least one 0 or $y = 0^r$ for some $r > 0$

But then for $i = 0$, $xy^i z = xz = 0^{n-r} 1^n \notin L$, a contradiction.

2nd non regular language

$L = \{0^p : p \text{ is prime}\}$ is not a regular language

Suppose L was regular

Let $n \geq 1$ be the pumping length as in Pumping Lemma

Consider the string $w = 0^n$, n is a prime

Then $w \in L$ and $|w| = n \geq n$

Hence, by the pumping lemma, w can be rewritten/decomposed as

$w = xyz$, where $y \neq \epsilon$, $|xy| \leq n$, and $xy^iz \in L$ for each $i \geq 0$

Consider any decomposition of w as $w = xyz$, where $y \neq \epsilon$, $|xy| \leq n$, and $xy^iz \in L$ for each $i \geq 0$

Then $x = 0^q$ and $y = 0^r$ and $z = 0^s$ where $q, s \geq 0$ & $r > 0$

By Pumping Lemma $xy^iz = 0^q 0^{ir} 0^s \in L$ for each $i \geq 0$

$\Rightarrow q + ir + s$ is a prime for all $i \geq 0$

But, for $i = q + 2r + s + 2$,

$q + ir + s = (r + 1)(q + 2r + s)$, which is a product of two numbers each ≥ 1 , a contradiction

Non regular language

$L = \{w \in \{0, 1\}^* : \text{the number of 0's in } w \text{ equals the number of 1's in } w\}$

Observe

Exactly the same arguments works as in proving

$L' = \{0^n 1^n : n \geq 0\}$ is not a regular language

OR

Closure Properties can be used to prove language is not regular.

Suppose L was regular.

Note $L \cap 0^*1^* = L'$

And by *closure property* under *intersection* of regular languages, L' would a regular language.

A contradiction.

Non regular language

Let $L = \{ww : w \in \{0, 1\}^*\}$

Suppose L was regular

Let $n \geq 1$ be the pumping length as in Pumping Lemma

Consider the string $w = 0^n 1 0^n 1$

Then $w \in L$ and $|w| = 2n + 2 \geq n$

Hence, by the pumping lemma, w can be rewritten/decomposed as $w = xyz$, where $y \neq \epsilon$, $|xy| \leq n$, and $xy^i z \in L$ for each $i \geq 0$

Consider any decomposition of w as $w = xyz$, where $y \neq \epsilon$, $|xy| \leq n$, and $xy^i z \in L$ for each $i \geq 0$

Then $x = 0^q$ and $y = 0^r$ where $q \geq 0$ & $r > 0$

Let $i = 2$, then $xy^2 z = xyyz$ has more 0's before the first 1 than after 1.

Therefore $xy^2 z \notin L$. A contradiction.